Reasoning test

For admission to Class 12

About the test

- 15 questions in 15 minutes: 5 each of maths, English and science, mixed together.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers

The questions test reasoning, not memory, using ideas taught up to Class 11. Chapters are from the NCERT (CBSE) books a student joining Class 12 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Mathematics

- *Whole Numbers* – number patterns and series
- *Playing with Numbers* – primes, factors, HCF and LCM
- *Fractions*
- *Mensuration* – perimeter and area
- *Ratio and Proportion*

Class 7 · Mathematics

- *Fractions and Decimals* – fractions
- *Data Handling* – averages (mean)
- *Simple Equations*
- *Comparing Quantities* – percentages, profit and loss, discounts, interest; ratio and proportion
- *Perimeter and Area*

Class 7 · Science

- *Motion and Time* – speed, distance and time

Class 8 · Mathematics

- *Linear Equations in One Variable* – simple equations
- *Squares and Square Roots* – squares and cubes
- *Cubes and Cube Roots* – squares and cubes
- *Comparing Quantities* – percentages, profit and loss, discounts, interest

Class 10 · Mathematics

- *Arithmetic Progressions* – number patterns and series
- *Probability*

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures · Overlapping groups (Venn diagrams)

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Science

- *Fun with Magnets* – magnets
- *Light, Shadows and Reflections* – light, shadows and plane mirrors
- *Electricity and Circuits* – electric circuits

Class 7 · Science

- *Heat* – heat and conductors
- *Acids, Bases and Salts* – acids, bases and indicators
- *Physical and Chemical Changes* – physical and chemical changes, rusting
- *Nutrition in Plants* – photosynthesis and plant respiration
- *Respiration in Organisms* – respiration; photosynthesis and plant respiration
- *Motion and Time* – speed
- *Light* – light, shadows and plane mirrors

Class 8 · Science

- *Force and Pressure* – force and pressure, pressure in liquids
- *Sound* – sound, and the speeds of light and sound

Class 9 · Science

- *Matter in Our Surroundings* – evaporation and condensation; change of state
- *Is Matter Around Us Pure?* – elements, compounds, mixtures and solutions
- *Force and Laws of Motion* – inertia, action and reaction
- *Gravitation* – mass and weight; floating, sinking and density
- *Work and Energy* – energy changes

Class 10 · Science

- *Electricity* – electric power and safety
- *Our Environment* – food chains

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS

   Find the next number in the series: 1, 2, 6, 24, 120, ___

   A 240 **B** 600

   C 720 **D** 840

2. ENGLISH

   Choose the word most nearly opposite in meaning to "transparent".

   A opaque **B** clear

   C bright **D** thin

3. SCIENCE

   When a hanging carpet is beaten with a stick, dust falls out of it. This is mainly because

   A the dust stays put as the carpet moves **B** the stick attracts the dust towards it

   C the dust becomes heavier when it is hit **D** the air pushes the dust inwards

4. MATHS

   A bicycle bought for ₹4,000 is sold at a loss of 15%. What is the selling price, in rupees?

   Type-in answer: _____

5. ENGLISH

   The soldier was praised for his ___ in rescuing the trapped villagers.

   A cowardice **B** courage

   C carelessness **D** confusion

6. SCIENCE

   Why is it dangerous to operate an electric switch with wet hands?

   A Wet hands slip off the switch **B** Water lets current pass into the body

   C Water damages the switch's plastic **D** Wet skin makes the switch heat up fast

7. MATHS

   Which of these is a perfect square?

   A 1,024 **B** 1,210

   C 1,440 **D** 1,690

8. ENGLISH

   Choose the correct words: By the time we reached the station, the train ___.

   A has left **B** leaves

   C is leaving **D** had left

9. SCIENCE

   An electric kettle draws a current of 5 A from a 230 V supply. What is its power, in watts?

   Type-in answer: _____

10. MATHS

A town's population of 20,000 grows by 10% every year. What will it be after 2 years?

A 24,000
B 24,200
C 24,400
D 26,000

11. ENGLISH

"Book the tickets today; the fares go up next week." This advice assumes that

A fares will never come down again
B the train is already nearly full
C the listener still plans to go
D next week is a public holiday

12. SCIENCE

On a humid day, drops of water form on the outside of a glass of iced water. This water comes from

A water seeping out through the glass
B the glass giving off moisture
C water vapour in the air condensing
D ice melting inside the glass

13. MATHS

One card is drawn at random from a well-shuffled pack of 52. What is the probability that it is a king or a heart?

A 1/13
B 1/4
C 17/52
D 4/13

14. ENGLISH

Pointing to a young man, Neha said, "His mother is the only daughter of my mother." How is the man related to Neha?

A Brother
B Son
C Nephew
D Cousin

15. SCIENCE

Lights in a house are wired in parallel, not in series. The main reason is that in parallel

A the house draws a smaller total current
B each bulb glows dimly and saves energy
C one switch turns every bulb on together
D each bulb gets full voltage on its own

Answers

1	**C. 720**	Each number is the one before multiplied by 2, then 3, then 4, then 5. The next step is × 6, and 120 × 6 = 720.
2	**A. opaque**	Transparent means light passes through so you can see through it; opaque means light does not pass through. "Clear" means the same as transparent.
3	**A. the dust stays put as the carpet moves**	The carpet is suddenly pushed away, but the dust tends to stay at rest (inertia), so it is left behind and falls out.
4	**3400**	15% of ₹4,000 is ₹600, so the bicycle sells for 4,000 − 600 = ₹3,400. This is a type-in question.
5	**B. courage**	He was praised, so the missing word must be a good quality shown during a rescue. Only courage fits; the rest would be criticised.
6	**B. Water lets current pass into the body**	Ordinary water conducts electricity, so moisture can carry current from the switch into the body and cause a shock.
7	**A. 1,024**	1,024 = 32 × 32. The others fall between squares: 34^2 = 1,156 and 35^2 = 1,225; 37^2 = 1,369 and 38^2 = 1,444; 41^2 = 1,681 and 42^2 = 1,764.
8	**D. had left**	Two past events are compared, and the train left first. The earlier of two past actions takes the past perfect: "had left".
9	**1150**	Power is voltage × current: 230 V × 5 A = 1,150 W. This is a type-in question: the answer is typed as a number.
10	**B. 24,200**	After one year: 22,000. The second year's 10% is taken on 22,000, adding 2,200, so 24,200. 24,000 forgets the growth compounds.
11	**C. the listener still plans to go**	Advice to book early only makes sense if the listener is going to travel. The other statements are not needed for the advice to hold.
12	**C. water vapour in the air condensing**	The cold glass cools the air touching it, and the water vapour in that air condenses into drops. Glass does not let water through.
13	**D. 4/13**	There are 4 kings and 13 hearts, but the king of hearts is in both, so 16 cards qualify: 16/52 = 4/13. Counting it twice gives 17/52.
14	**B. Son**	The only daughter of Neha's mother is Neha herself. So the man's mother is Neha, and he is her son. "Nephew" assumes a sister.
15	**D. each bulb gets full voltage on its own**	In parallel every bulb gets the full mains voltage and has its own path, so each can be switched on or off without affecting the others.